

Armaan Amatya

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EDUCATION

University of Houston

Bachelor of Science in Computer Science and Mathematics, GPA: 3.5

Houston, TX

Aug. 2024 – Dec. 2026

EXPERIENCE

Autonomize AI

Austin, TX

Machine Learning Engineer Intern

May 2026 – Aug. 2026

- Quantized Qwen2.5-VL-32B (4-bit AWQ) and GRPO post-trained it for healthcare OCR at 95% of full-model field accuracy on 1,000+ documents; served via vLLM as an HTTP endpoint on a 16GB Azure T4 with CPU offload
- Made production OCR on Azure and GCP 35% faster at under 2% accuracy loss by pruning 50%+ redundant document-image tokens, cutting KV-cache memory 40%
- Integrated a model router into the platform to send pages to smaller, cheaper OCR models
- Shipped a RAG chatbot over 1,000+ policy docs, saving 20+ hours/week at 90%+ accuracy and 2s per question

University of Houston

Houston, TX

Undergraduate Researcher

Jan. 2026 – Present

- Reduced multimodal LLM inference latency 40% at 98% of baseline accuracy with a novel token compression method that cuts visual tokens 60% and the KV-cache footprint on Qwen2.5/3-Omni; won Best Research Award
- Deployed compressed multimodal models to NVIDIA Jetson Orin for on-device inference, trained with multi-GPU distributed PyTorch (DDP and FSDP) on RTX 6000s

Artinafti

Remote

AI Engineer Intern

Jan. 2026 – May 2026

- Beat 10+ open-source upscalers on 2k+ blurry test images by fine-tuning Real-ESRGAN on 4,000 artwork pairs
- Improved image quality 25% (FID) by LoRA fine-tuning Flux on 5k+ artworks, with color-tone controls
- Cut face and text warping in generation with CodeFormer restoration, text preservation and Flux ControlNet guidance
- Served ESRGAN and 8-bit Flux at 1,000+ requests a day on an AWS A10G, with a Redis job queue and cache and results in S3; containerized with Docker and deployed by GitHub Actions, monitored in Sentry

AutoHDR

Austin, TX

Machine Learning Engineer Intern

Aug. 2025 – Dec. 2025

- Cut per-photo enhancement from 5 minutes to under 15 seconds on 100,000+ real-estate listing photos a month with PyTorch exposure fusion, denoising and 4x super-resolution, deployed as a FastAPI service on AWS EC2
- Cut redundant processing 25% with an image deduplication service, exposed by FastAPI to the TypeScript backend
- Produced 1,000+ real-estate tour videos monthly with a depth-aware image-to-video model deployed via FastAPI

FuseMachines

New York, NY

Machine Learning Engineer Intern

May 2024 – Aug. 2024

- Fine-tuned FaceNet to 98% precision for a production attendance app, then served it from a Dockerized FastAPI service, with ONNX to FP16 TensorRT cutting latency and cloud cost 40%

PROJECTS

Mixed-Precision GEMM Kernels in Triton | [Link](#) | Triton, CUDA, PyTorch, Nsight Compute

- Wrote an FP16 GEMM kernel in Triton that hit 104% of cuBLAS on Blackwell and 98% on Ampere at 8192x8192
- Took a fixed-tile kernel from 46-69% to 98-104% of cuBLAS via 3-stage pipelining and a 192-config autotune sweep

Facial Recognition on AMD Kria KV260 SoC | [Link](#) | C++, Vitis AI, Vivado, GStreamer, INT8

- Recognized faces on-device at 99.47% accuracy and 98% TAR at 0.05% FAR on a 1k+ image held-out set
- Ran at 30 FPS, 100x the Cortex-A53 CPUs, via a hardware-accelerated zero-copy DMA GStreamer pipeline

Domain QA System for Scientific PDFs | [Link](#) | HuggingFace, PEFT, BitsAndBytes, FAISS, LangChain

- QLoRA-tuned Mistral-7B on 234 QA pairs with hybrid RAG over 75 papers, 80% GPT-4o-as-judge win rate

OpenResearch | [Link](#) | Python, Harbor, PaperBench, Kubernetes, GCP

- Engineered an RLM-inspired agent harness that reproduces arXiv papers autonomously, with sub-agents coding each hypothesis in an isolated branch and sandbox, and PyTorch DDP and FSDP built in
- Faithfully reproduced the SDAR agentic-RL paper, training Qwen 1.7B, 3B and 7B across 4 GPUs

TECHNICAL SKILLS

Languages: Python, C++, CUDA, Triton, TypeScript, JavaScript, Go, Rust, SQL, Bash

Frameworks: PyTorch, vLLM, TensorRT, HuggingFace, scikit-learn, LangChain, DSPy, FastAPI, React, Next.js, Node.js

Cloud: AWS, EC2, S3, Azure, GCP, Docker, Kubernetes, CI/CD, MLOps, PostgreSQL, Redis, Sentry, PostHog

AI/ML: LLMs & model architectures, transformers & attention, RL (PPO/GRPO/RLHF), post-training, LoRA/QLoRA fine-tuning, AWQ quantization, inference optimization, RAG, FAISS, LLM Agents, MCP, Evals, ComfyUI